Maths foundations: worked answers

20 original questions for 11 Plus skill preparation. Supports families preparing for tests taken in 2027; not an official paper, past paper or school-specific mock.

Scope: order of operations; factors, multiples and primes; negative numbers; percentage change.

Use untimed first. Check the method as well as the final answer. Return to the linked guide for teaching and further questions.

Find both booklets and online practice: https://www.11plusexampapers.com/papers

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Order of Operations

1. Work out 14 + 6 × 5.

Answer: 44

Why: Multiply 6 × 5 = 30, then add 14. The answer 100 would come from adding first.

2. Work out 48 ÷ 8 × 4.

Answer: 24

Why: Division and multiplication have equal priority: 48 ÷ 8 = 6, then 6 × 4 = 24.

3. Work out 25 − 9 + 2.

Answer: 18

Why: Subtraction and addition share a level. Read left to right: 25 − 9 = 16, then 16 + 2 = 18.

4. Work out 7 × (13 − 8) + 6.

Answer: 41

Why: The bracket gives 5. Multiply 7 × 5 = 35, then add 6.

5. Work out 4^2 + 18 ÷ 6.

Answer: 19

Why: 4^2 = 16 and 18 ÷ 6 = 3. Add them to get 19.

Source guide: https://www.11plusexampapers.com/blog/11-plus-order-of-operations

Factors, Multiples and Primes

6. List every positive factor of 30.

Answer: 1, 2, 3, 5, 6, 10, 15, 30

Why: The pairs are 1 × 30, 2 × 15, 3 × 10 and 5 × 6. Write the list in increasing order.

7. Which are prime: 1, 2, 21, 29, 39?

Answer: 2 and 29

Why: 1 has only one factor; 21 = 3 × 7 and 39 = 3 × 13. Only 2 and 29 have exactly two positive factors.

8. Find the highest common factor of 24 and 40.

Answer: 8

Why: The common factors are 1, 2, 4 and 8. The greatest is 8, not 40, which does not divide 24.

9. Find the lowest common multiple of 9 and 12.

Answer: 36

Why: Positive multiples of 12 are 12, 24, 36. Of these, 36 is the first divisible by 9.

10. There are 42 pencils and 70 stickers. Make the greatest number of identical packs using everything. How many packs, and what goes in each?

Answer: 14 packs; 3 pencils and 5 stickers each

Why: HCF(42, 70) = 14. Divide each original quantity by 14 to find the contents.

Source guide: https://www.11plusexampapers.com/blog/11-plus-factors-multiples-primes

Negative Numbers

11. Put −3, 5, −11, 0 and −1 in ascending order.

Answer: −11, −3, −1, 0, 5

Why: Increasing order means moving right along the number line, starting with the smallest signed value.

12. Which is greater, −8 or −5?

Answer: −5

Why: −5 is further right and closer to zero. Compare the signed numbers, not just their digits.

13. A temperature of −6°C rises by 11°C. What is the final reading?

Answer: 5°C

Why: Six degrees of the rise reach zero. The remaining five degrees take it to 5°C.

14. A reading is 7°C, then falls by 15°C. What is it now?

Answer: −8°C

Why: A fall of 7 reaches zero; a further fall of 8 reaches −8°C.

15. How many degrees warmer is 3°C than −9°C?

Answer: 12°C

Why: The interval is 9 degrees from −9 to zero plus 3 from zero to 3.

Source guide: https://www.11plusexampapers.com/blog/11-plus-negative-numbers

Percentage Increase and Decrease

16. Increase 180 by 15%.

Answer: 207

Why: 10% is 18 and 5% is 9. Add the increase of 27 to 180.

17. Reduce £72 by 25%.

Answer: £54

Why: The saving is £18, one quarter of £72. Subtract it from £72.

18. A £90 coat is reduced by 12%. How much money is saved?

Answer: £10.80

Why: 10% is £9 and 2% is £1.80. The question wants the saving, not the £79.20 sale price.

19. A £45 kit rises in price by 20%, then falls by 10% of the increased price. Find its final price.

Answer: £48.60

Why: The increase makes £54. Ten per cent of £54 is £5.40, leaving £48.60.

20. After a 20% reduction, a lamp costs £56. What was its original price?

Answer: £70

Why: £56 is 80%. Divide by 8 for 10% = £7, then multiply by 10. Check: £70 less £14 = £56.

Source guide: https://www.11plusexampapers.com/blog/11-plus-percentage-increase-decrease